Today's agenda

- ↳ float and double
- ↳ operators
- ↳ if else

AlgoPrep

// Decimal → 2.2, 1.3, 10.0, 3.76 etc.

↳ float & double

int → (.6-7 decimal Places)

long → (.14-15 decimal Places)

→ In integer, default values are int.

→ In decimal, default values are double.

Integer

↳ long l = 10^10 L;

Decimal

↳ float f = 2.4f; → write f at the end while initializing float.

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* 2 Golden rules of typecasting

↳ 1. if there is guaranted no loss of data: implicit

Ex: int to long.

float to double.

↳ 2. if there is a chance for loss of data then:

we can still do the conversion forcefully: explicit

Ex: long to int

double to float

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Quiz 1:

```
double d = 2.8;
System.out.println(d);
```

→ 2.8

Quiz 2:

```
float f = 3.3f;
System.out.println(f);
```

→ 3.3

Quiz 3:

```
float f = 3.4f;
double d = f;
System.out.println(d);
```

→ 3.4

3.4	3.4
d	f

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Quiz 4:

3.4
d

```
double d = 3.4;
float f = d;
System.out.println(f);
```

→ error

```
double d = 3.4;
float f = (float)d;
System.out.println(f);
```

← Correct Syntax

//operation

↳ Rule 1: Mathematical operation between decimal and non-decimal, Result: decimal

↳ Rule 2: Operation between same category but different capacity, Result: bigger size

Ex: int + long → long

float + double → double

long + double → double

int + int → int

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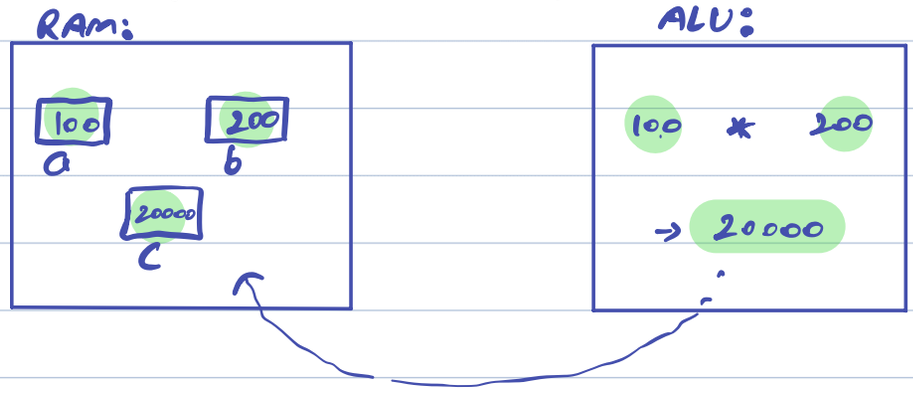
// Ex:

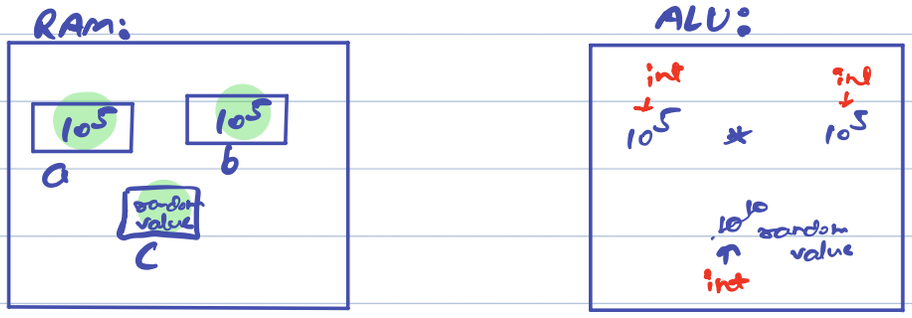
```
int a = 100;
int b = 200;
int c = a*b;
System.out.println(c);
```

→ 20000



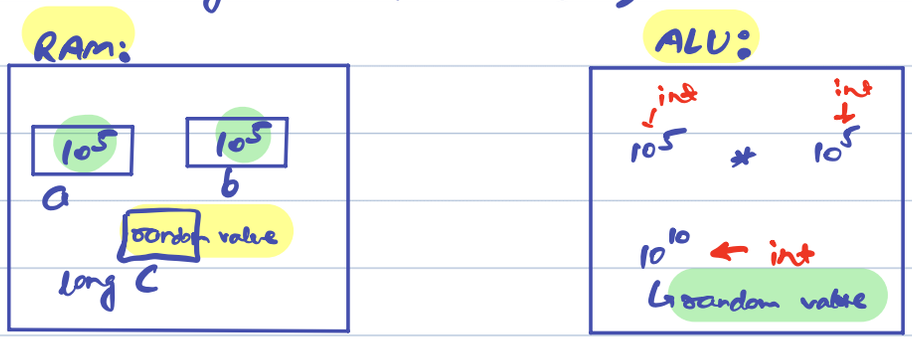
Quiz 5:

```
int a = 100000;
int b = 100000;        → random value
int c = a*b;
System.out.println(c);
```



Quiz 6:

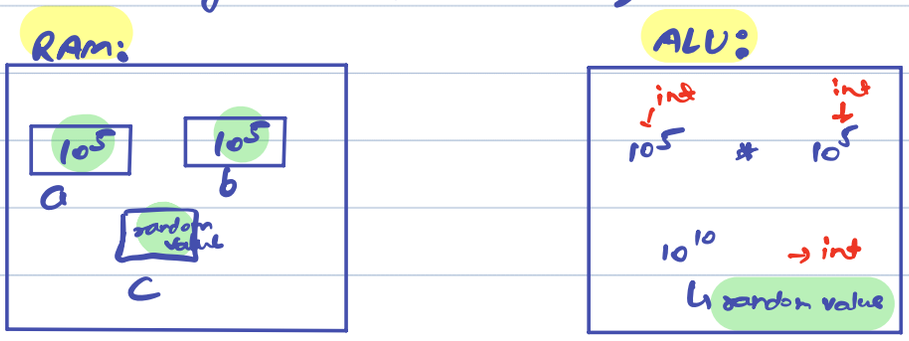
```
int a = 100000;
int b = 100000;        → random value
long c = a*b;
System.out.println(c);
```



Quiz 7:

```
int a = 100000;
int b = 100000;
long c = (long)(a * b);
System.out.println(c);
```

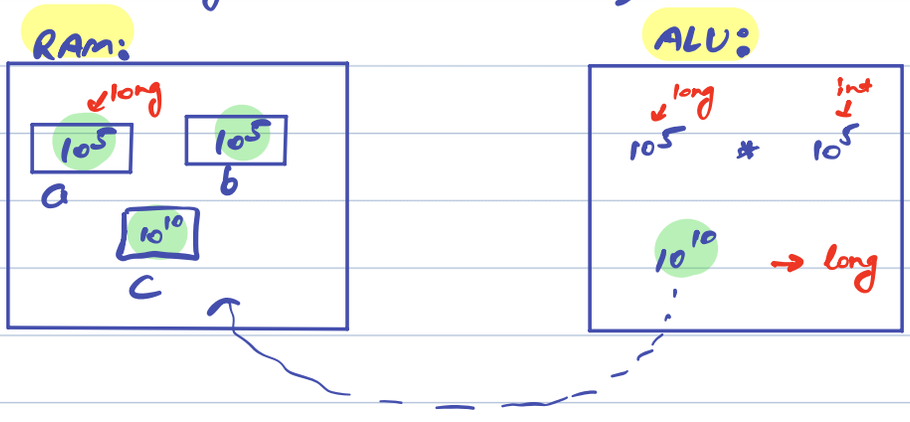
typecast → Calculation → random value



Quiz 8:

```
int a = 100000;
int b = 100000;
long c = (long)(a) * b
System.out.println(c);
```

→ 10^{10}



Break till 9:31 PM

* Arithmetic operators

↳ +, −, *, /, % ← remainder

↳ 20 % 3 = 2

↳ 50 % 10 = 0

Quiz 9:

System.out.println(16/3); → 5

ALU:

16 (int) / 3 (int)

~~5.333~~ → int

↳ 5

Quiz 10:

System.out.println(25.0/3);

↳ 8.333

ALU

25.0 (double) / 3 (int)

8.333... → double

Quiz 11:

System.out.println(35 % 9); → 8

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* **Relational operators** → used to check relation between 2 data.

ex: >, >=, ==, <etc.

	x=8 y=10	x=15 y=7	x=13 y=13
x less than y: x<y	true	false	false
x greater than y: x>y	false	true	false
x greater than equal to y: x>=y	false	true	true
x smaller than equal to y: x<=y	true	false	true
x equal y: x==y	false	false	true
x not equal y x!=y	true	true	false

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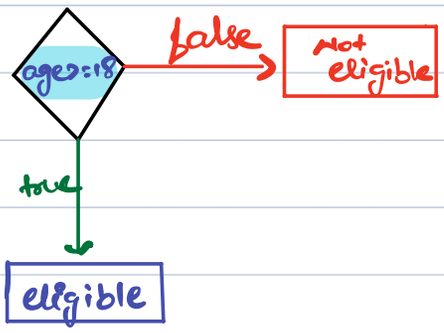
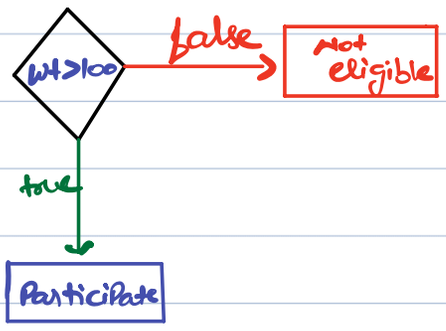
Conditional Statement

// If

1. Check if Person is eligible for car driving license.



2. if Person is above 100 kg in weight, he can Participate then only.



idea: when we want to do something on the basis of Condition being true.

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Syntax:

only cond^n that will return true / false can be written.

```
if (cond^n) {
    // lines you want to execute
    if the cond^n is true.
}
```

a) 4 == 5 ✓ (valid cond^n but will give you false)

b) 4 < 5 ✓✓

c) 4+5 ✗✗

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Q) Read a number (age of person), if person is eligible to get driving license Print "eligible" otherwise don't do anything.

```
Scanner scn = new Scanner(System.in);
int age = scn.nextInt();
if (age >= 18) {
    System.out.println("eligible");
}
```

Quiz 12:

```
int x=20;
if (x>=15){
    S.O.p("Hello");
}
S.O.p("Hello");
```

Hello Hello

20
x

Quiz 13:

```
int x=20;
int y=25;

if (x>=25){
    S.O.p("AlgoPrep1");
}
if (y>=25){
    S.O.p(AlgoPrep2);
}
```

AlgoPrep2

Quiz 14:

```
if (10>6){
    S.O.p("AlgoPrep1");
}
if (15 > 25){
    S.O.p("2nd");
}
```

AlgoPrep1

AlgoPrep

Quiz 15:

```
int x = 55;
int y = 65;

if (x > 55) {
    S.O.P ("first");
    x = x + 2;
}

if (y >= 60) {
    S.O.P ("second");
    y = y + 2;
}

S.O.P (x+y);
```

false

true

x	y
55	67

Second 122

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